3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

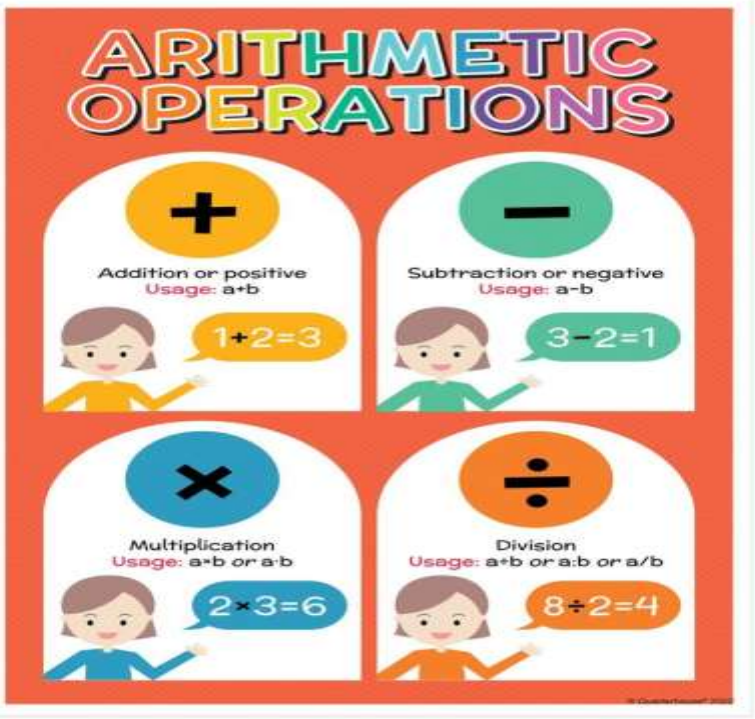
Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 18: Operators in Python

T FOUNDATION



Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what operators are.
- ✅ Perform arithmetic calculations using Python.
- ✅ Compare numbers using comparison operators.
- ✅ Use logical operators to combine conditions.
- ✅ Understand assignment operators.
- ✅ Write simple programs using operators.

3DWEBSOFT FOUNDATION

What are Operators?

An **operator** is a special symbol that tells the computer to perform an operation on values or variables.

For example:

- **+** means Add
- **-** means Subtract
- **==** means Compare
- **>** means Greater than

Real-Life Example

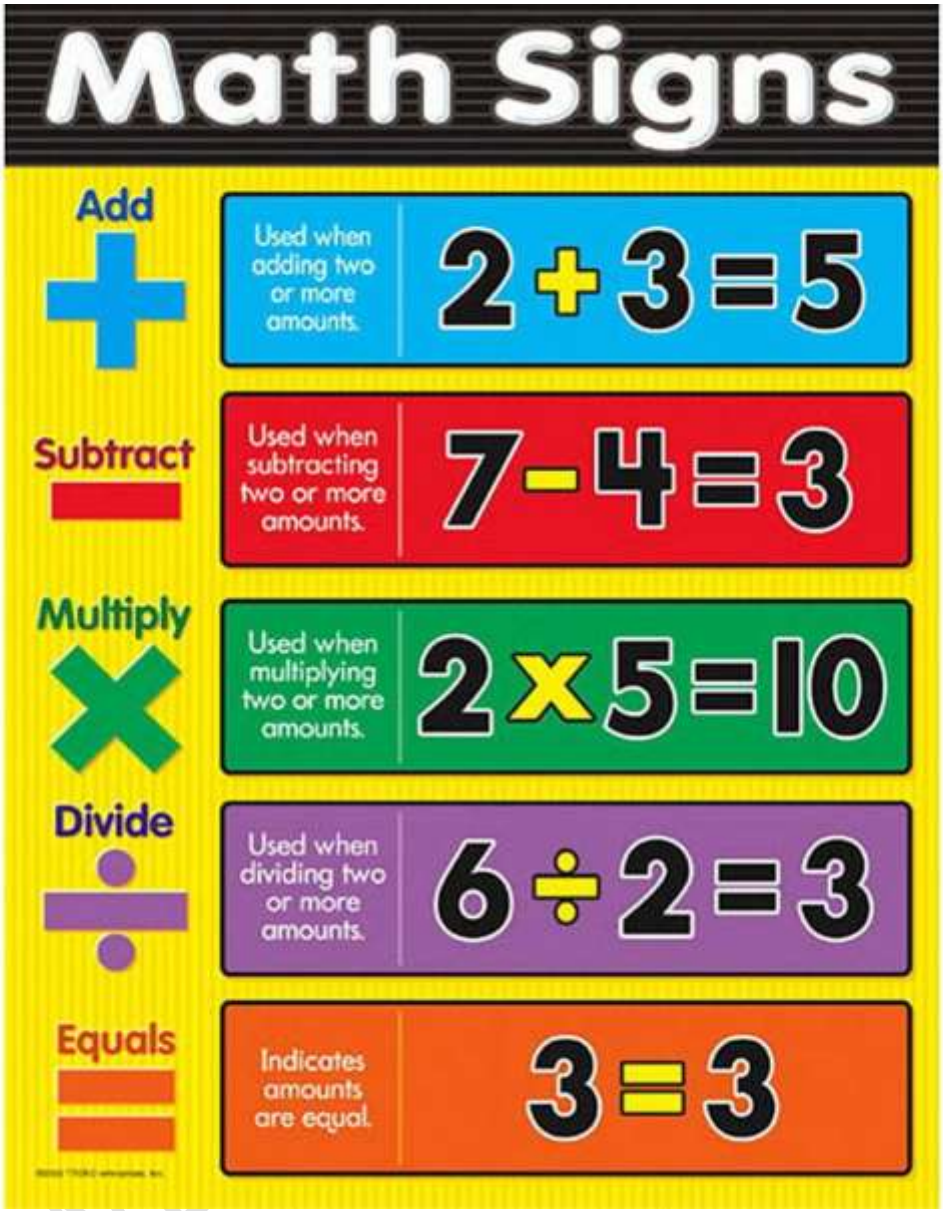
Suppose you have **5 chocolates** and your friend gives you **3 more**.

5 + 3 = 8

Here:

- **5** and **3** are **operands**
- **+** is the **operator**

What is an Operator?



PYTHON OPERATORS CHEAT SHEET

❶ ARITHMETIC

Operator	Example
+	5 + 3 = 8
-	5 - 3 = 15
*	5 / 2 = 2.5
/	5 // 2 = 2
%	5 % = 1
**	2 * 3 = 8

❷ ASSIGNMENT

Operator	Example
=	x = 5
+	x ± 3
+=	x ± 3

❸ COMPARISON

Operator	Example
==	5 == 5
!=	5 != 3
><	5 < 3
<=	5 <= 3

❹ LOGICAL

Operator	Example
and	(5 > 3) and (5 < 10)
or	not (5 > 3)

❺ BITWISE

& ⁝≡ 8	6 & 3 = 2
I	6 I 2 ¦ = 9
^	6 ^ 3 = 5
~x	~x = ~1
<< 2	6 << 2 = 24
>> 2	6 >> 2 = 1

❻ IDENTITY

Operator	Example
is	x is y
is not	x is not z

© A thinker by **Ashwin**

Types of Operators in Python

Operator Type	Purpose	Example
Arithmetic Operators	Perform calculations	+, -, *, /
Comparison Operators	Compare values	==, >, <
Assignment Operators	Assign values	=, +=
Logical Operators	Combine conditions	and, or, not

1 Arithmetic Operators

Arithmetic operators are used for mathematical calculations.

Operator	Meaning	Example
+	Addition	5 + 3
-	Subtraction	8 - 2
*	Multiplication	4 * 5
/	Division	10 / 2
%	Modulus (Remainder)	10 % 3
//	Floor Division	10 // 3
**	Exponent (Power)	2 ** 3

Addition (+)

a = 10

b = 5

3DWEBSOFT FOUNDATION

print(a + b)

Output

15

Subtraction (-)

a = 20

b = 8

print(a - b)

Output

12

Multiplication (*)

a = 6

b = 4

print(a * b)

Output

24

Division (/)

a = 12

b = 3

3DWEBSOFT FOUNDATION

print(a / b)

Output

4.0

Modulus (%)

Gives the **remainder** after division.

print(10 % 3)

Output

1

Floor Division (//)

Removes the decimal part.

print(10 // 3)

Output

3

Exponent (**)

Means "power of".

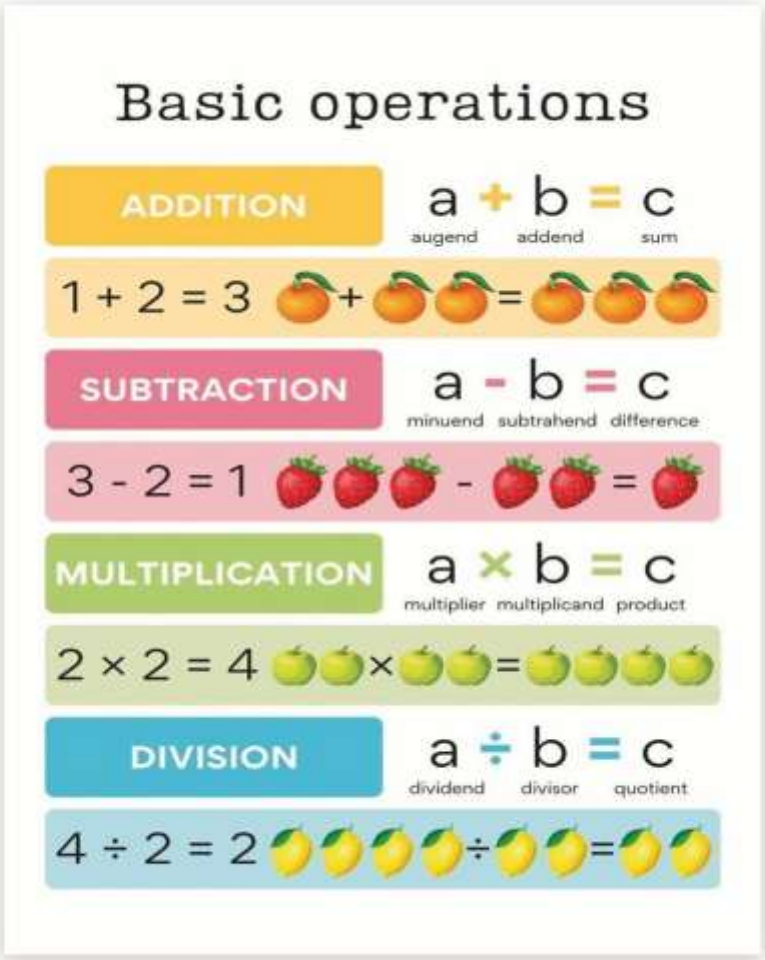
print(2 ** 3)

Output

8

Arithmetic Operators

3DWEBSOFT FOUNDATION



Arithmetic Operators

Operator	Meaning	Example
+	Addition	4 + 7 ⟶ 11
-	Subtraction	12 - 5 ⟶ 7
*	Multiplication	6 * 6 ⟶ 36
/	Division	30 / 5 ⟶ 6
%	Modulus	10 % 4 ⟶ 2
//	Quotient	18 // 5 ⟶ 3
**	Exponent	3 ** 5 ⟶ 243



2 Comparison Operators

Comparison operators compare two values.

They always return:

- **True**
- **False**

Operator	Meaning
==	Equal
!=	Not Equal

Operator	Meaning
>	Greater Than
<	Less Than
>=	Greater Than or Equal
<=	Less Than or Equal

Example

```
print(10 == 10)
```

Output

True

```
print(5 > 8)
```

Output

False

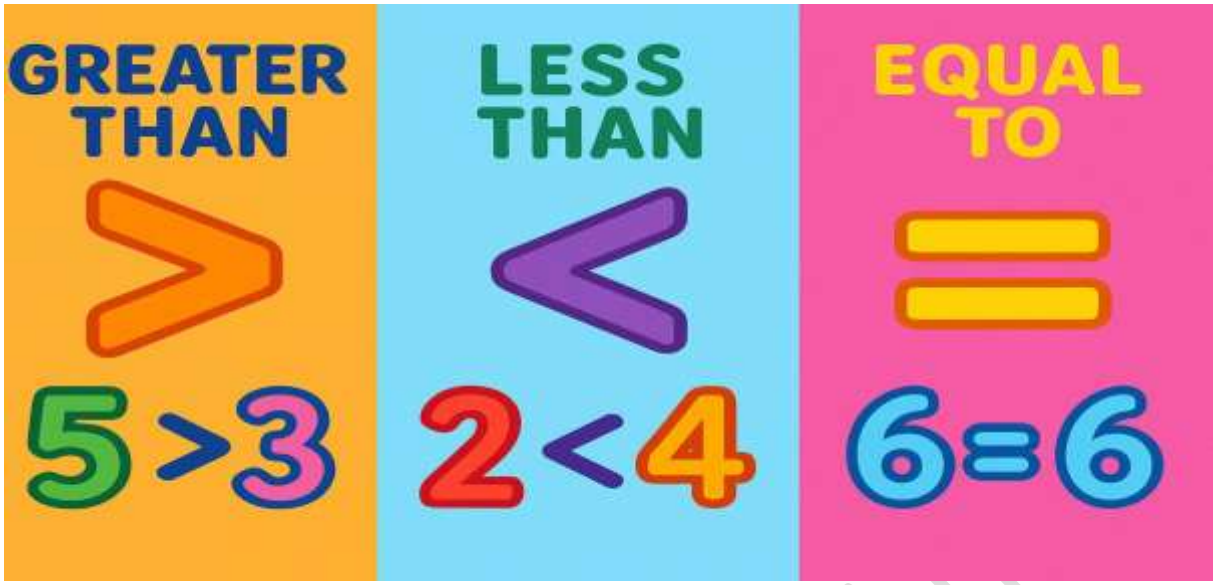
```
print(20 != 15)
```

Output

True

Comparison Operators

3DWEBSOFT FOUNDATION



Comparison Operators

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

3 Assignment Operators

Assignment operators store values in variables.

Operator	Meaning
=	Assign
+=	Add and Assign
-=	Subtract and Assign
*=	Multiply and Assign
/=	Divide and Assign

Example

```
x = 5


print(x)
```

Output

```
5
```

Using +=

```
x = 5


x += 3


print(x)
```

Output

```
8
```

3DWEBSOFT FOUNDATION

Using -=

x = 10

x -= 4

print(x)

Output

6

4️⃣ Logical Operators

Logical operators combine conditions.

Operator	Meaning
and	Both conditions must be True
or	At least one condition is True
not	Opposite result

AND Example

print(10 > 5 and 8 > 4)

Output

True

OR Example

print(10 > 15 or 8 > 4)

3DWEBSOFT FOUNDATION

Output

True

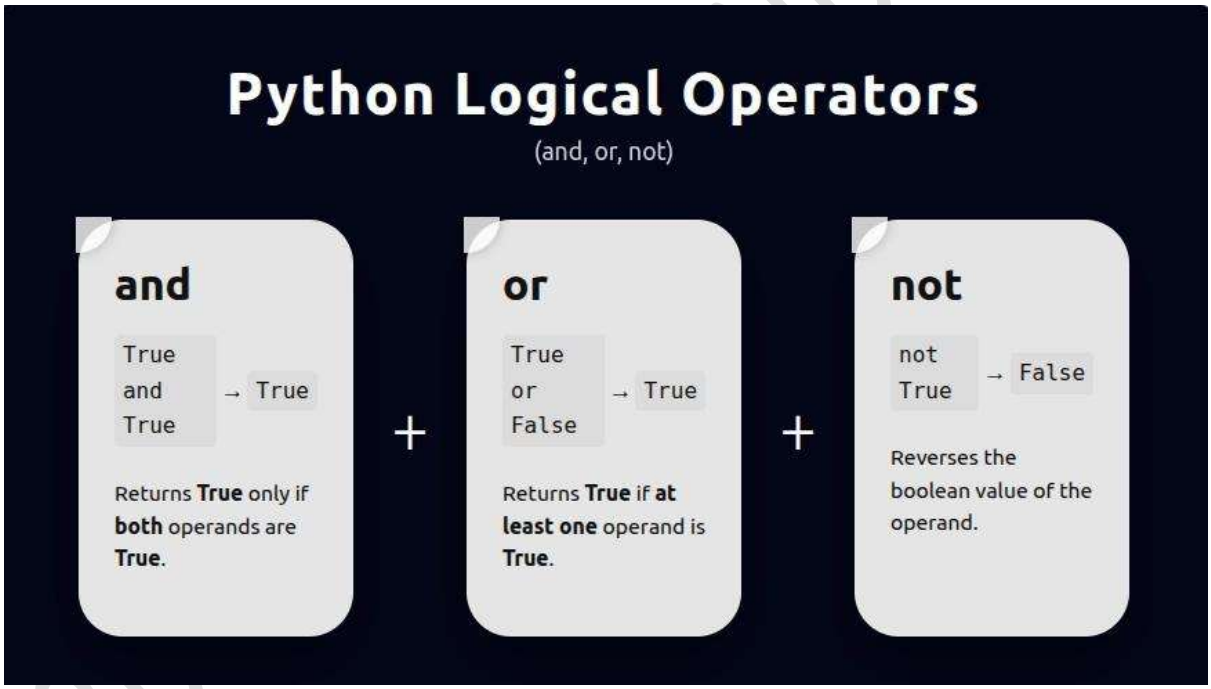
NOT Example

print(not(5 > 2))

Output

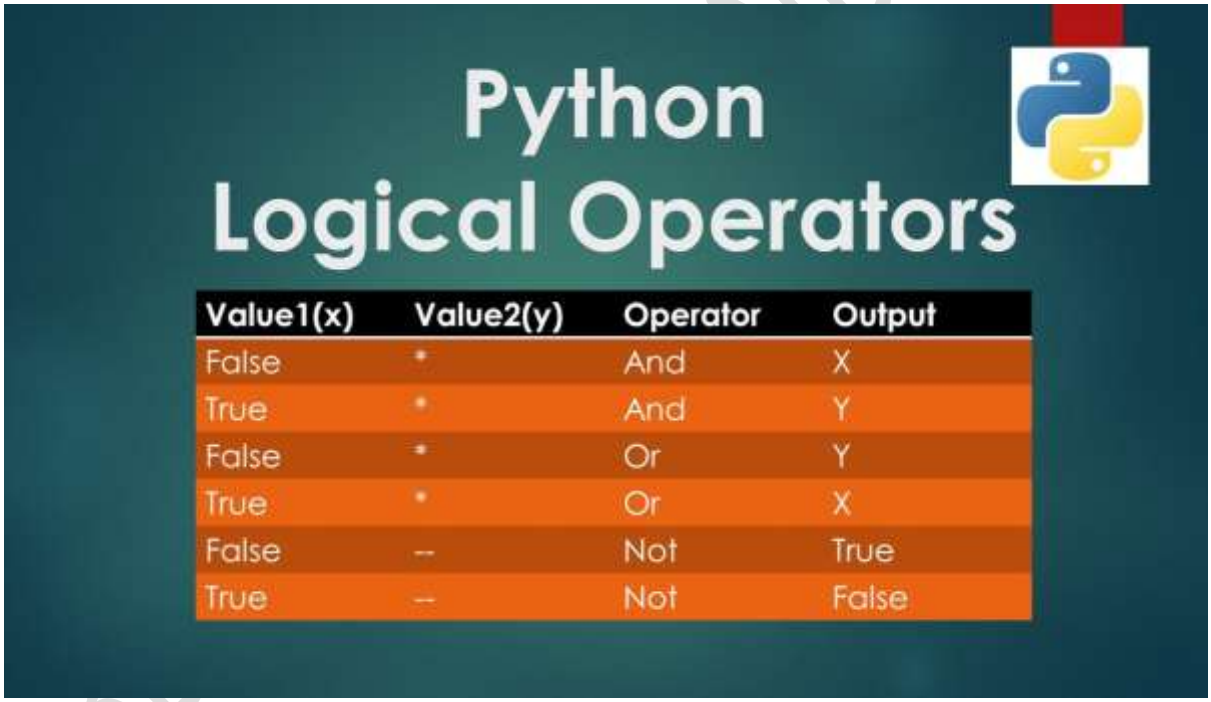
False

Logical Operators



LOGICAL OPERATORS

logical operator	Description	Syntax
and	True if both statements are true	a and b
or	True if one of the statements is true	a or b
not	Reverse the result. If the result is true it will return false and viceversa	not a not b



Order of Operations (PEMDAS)

Python follows this order:

1. Parentheses ( )
2. Exponent **

3. Multiplication * and Division /
4. Addition + and Subtraction -

Example

```
print((2 + 3) * 4)
```

Output

20

Practical Session

Practical Activity 1

Addition Program

```
a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("Sum =", a + b)
```

Practical Activity 2

Subtraction Program

```
a = int(input("Enter Number: "))
b = int(input("Enter Number: "))

print("Difference =", a - b)
```

Practical Activity 3

Multiplication

3DWEBSOFT FOUNDATION

```
a = int(input("Enter Number: "))
b = int(input("Enter Number: "))

print("Product =", a * b)
```

Practical Activity 4

Division

```
a = int(input("Enter Number: "))
b = int(input("Enter Number: "))

print("Quotient =", a / b)
```

Practical Activity 5

Compare Two Numbers

```
a = int(input("First Number: "))
b = int(input("Second Number: "))

print(a == b)
print(a > b)
print(a < b)
```

Practical Activity 6

Find Remainder

```
number = int(input("Enter Number: "))
```

3DWEBSOFT FOUNDATION

```
print(number % 2)
```

Practical Activity 7

Square of a Number

```
num = int(input("Enter Number: "))


print(num ** 2)
```

Practical Activity 8

Update Marks

```
marks = 80


marks += 10


print(marks)
```

Classroom Activities

Activity 1

Calculate:

- 25 + 15
- 50 - 18
- 12 × 6

3DWEBSOFT FOUNDATION

- 81 ÷ 9

Activity 2

Predict the output.

```
print(15 > 8)


print(7 == 10)


print(12 != 5)
```

Activity 3

Write a program to calculate the area of a rectangle.

Formula:

Area = Length × Breadth

Program

```
length = int(input("Length: "))
breadth = int(input("Breadth: "))

print("Area =", length * breadth)
```

Activity 4

Write a program to calculate the perimeter of a rectangle.

Formula:

Perimeter = 2 × (Length + Breadth)

3DWEBSOFT FOUNDATION

Real-Life Applications

Situation	Operator Used
Shopping Bill	+
Temperature Difference	-
School Marks Calculation	+
Comparing Exam Scores	>
Password Checking	==
Sharing Chocolates	/

Common Mistakes

❌ Using one = instead of ==

Wrong

```
print(5 = 5)
```

✅ Correct

```
print(5 == 5)
```

❌ Forgetting int() when taking numbers as input

Wrong

```
a = input()
b = input()

print(a + b)
```

3DWEBSOFT FOUNDATION

If you enter:

10

20

Output:

1020

Correct

```
a = int(input())
b = int(input())

print(a + b)
```

Output

30

Keywords

- **Operator** – A symbol that performs an operation.
- **Operand** – The value on which an operator works.
- **Arithmetic Operator** – Performs mathematical calculations.
- **Comparison Operator** – Compares two values.
- **Assignment Operator** – Assigns values to variables.
- **Logical Operator** – Combines multiple conditions.

Lesson Summary

- Operators perform actions on values and variables.

3DWEBSOFT FOUNDATION

- Arithmetic operators are used for calculations.
- Comparison operators return **True** or **False**.
- Assignment operators store and update values.
- Logical operators combine conditions.
- Operators help us solve real-life mathematical and logical problems in Python.

Quiz

Multiple Choice Questions

1. Which operator is used for addition?

A) -

B) +

C) *

D) /

✅ **Answer:** B

2. Which operator checks equality?

A) =

B) ==

C) !=

D) >=

✅ **Answer:** B

3. Which operator gives the remainder?

3DWEBSOFT FOUNDATION

A) %

B) /

C) *

D) +

✅ **Answer:** A

4. What is the output?

print(4 * 5)

A) 9

B) 20

C) 25

D) 45

✅ **Answer:** B

5. What is the output?

print(10 > 15)

A) True

B) False

C) 10

D) Error

✅ **Answer:** B

Homework

3DWEBSOFT FOUNDATION

1. Write a program to add two numbers.
2. Write a program to subtract two numbers.
3. Write a program to multiply two numbers.
4. Write a program to compare two numbers using > and ==.
5. Write a program to calculate the area of a rectangle.
6. Write a program to find the square of a number using **.
7. Write a program to find the remainder using %.

3DWEBSOFT FOUNDATION